Geometry
Unit 2
Lesson 1 Practice

Name____________________
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Period____________________

For questions 1-3, determine if each of the following represents a definition, postulate, or theorem. Then, draw a picture using the given information.

1. If M is the midpoint of a line segment AB, then $AM = \frac{1}{2}AB$ and $MB = \frac{1}{2}AB$.

2. Vertical angles are congruent.

3. Perpendicular lines are lines that intersect to form a right angle.

Complete the table.

	Property	**Equality**	**Congruence**
4.	Transitive	If $m\angle ABC = m\angle DEF$ and $m\angle DEF = m\angle DEF = 45°$, then $m\angle ABC = 45°$	If $\angle GHI \cong \angle FED$ and $\angle FED \cong \angle ABC$, then ____________________
5.	Reflexive		$\angle FGH \cong \angle HGF$
6.		If $m\angle XYZ = 19°$,then $19° = m\angle XYZ$	If $\angle XYZ = \angle FGH$, then $\angle FGH = \angle XYZ$

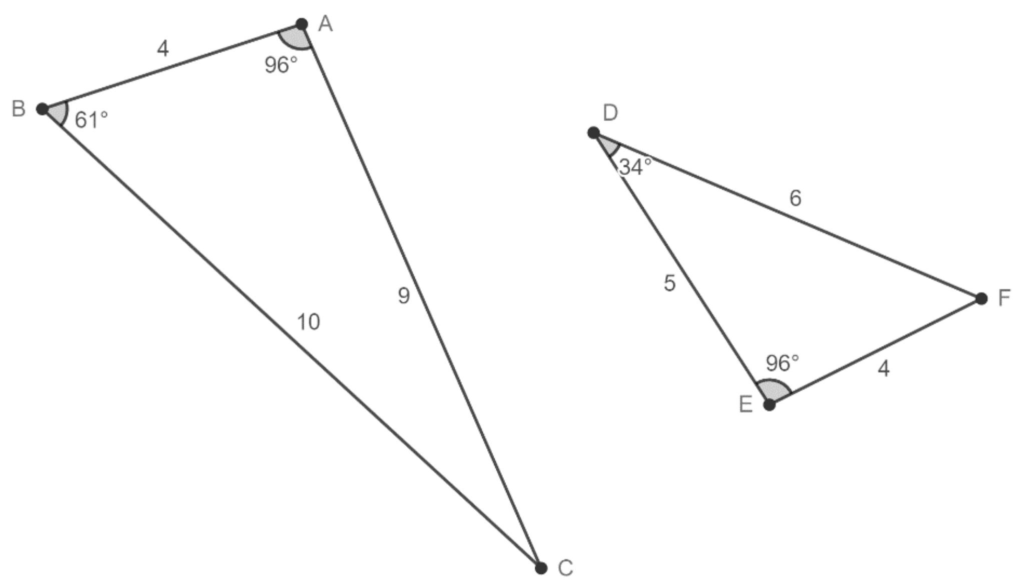
7. Write the conclusion for the statement and name the property.

If $m\angle QRS + m\angle LMN = 223°$ and $223° = m\angle GHI$, then…

Complete the table with the reason that makes the given statement true.

	Statement	Reason
8.	$m\angle MNP = m\angle DFG$	
9.	$\angle MNP \cong \angle DFG$	

10. Triangles ABC and DEF are shown.



Fill in the blanks using the information shown on the two triangles.

$EF =$ _______. Therefore, $\overline{EF} \cong$ _______.

$m\angle CAB =$ _______. Therefore, $\angle CAB \cong$ ________.

Are triangles ABC and DEF congruent? Explain.

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For questions 1-5, determine the type of reasoning for each statement.

1. For a golf cart to be considered 'street legal' it must have blinkers. Jill's golf cart has been deemed 'street legal'. Therefore, her golf cart has blinkers.

2. When an object is thrown up in the air, it comes back down. Mark throws a baseball up in the air. The baseball will come back down.

3. When Ethan stays up late, he is tired the next day. Ethan stayed up last night until $11{:}30$ pm, therefore, Ethan will be tired the next day.

4. The Olympic games are held every four years. The last games were held in 2020. The next games will be held in 2024.

5. If a student passes an entrance examination, they will be admitted into their local community college. Barry passed the entrance examination. Barry will be admitted into the community college.

For each statement in 6-10, write a counterexample that disproves the statement.

6. If a student attends the junior/senior prom, the student is either a junior or a senior.

7. If someone owns a cell phone, they are able to access the internet from it.

8. When two numbers are added, the result will always be a larger number.

9. The product of two odd numbers will be even.

10. If a number is divisible by 2, it is also divisible by 4.

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You and a partner are playing the popular game UNO. You realize there was an error made in one of the steps. Determine which series of steps the error was made in and circle it.

	Statement	**Reason**
1.	Red 9	Given
2.	Red Skip	Matches color
3.	Yellow Skip	Matches word
4.	Yellow 5	Matches color
5.	Blue 5	Matches number
6.	Green 7	Changes color to green

Complete the proof for the card game UNO.

	Statement	Reason
7.	Red 4	
8.		Changes color to yellow
9.	Yellow Draw Two	
10.	Yellow 8	

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For questions 1-3, use the quote shown.

"If you look at what you have in life, you'll always have more" – Oprah Winfrey

1. Write the converse of this statement.

2. Write the inverse of this statement.

3. Write the contrapositive of this statement.

For questions 4-6, use the quote shown.

"The harder I work, the more luck I seem to have" – Thomas Jefferson

4. Write the converse of this statement.

5. Write the inverse of this statement.

6. Write the contrapositive of this statement.

For questions 7-8, use the quote shown.

“If life were predictable it would cease to be life, and be without flavor” – Eleanor Roosevelt

7. What is the hypothesis?

8. What is the conclusion?

For questions 9-10, use the quote shown.

“If you’re not making mistakes, then you’re not doing anything”- John Wooden

9. Write the inverse.

10. Write the contrapositive of the quote.

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Lesson 5 Practice Period____________________

Complete the table for each given conditional statement.

If a shape is a triangle, then it has exactly three sides.			
	Relationship to Conditional	**Statement**	**True or False**
1.	Converse		
2.	Inverse		
3.	Contrapositive		
Write the conditional using "if and only if".			
4.			

Two angles are supplementary if their sum is $180°$.			
	Relationship to Conditional	**Statement**	**True or False**
5.	Converse		
6.	Inverse		
7.	Contrapositive		
Write the conditional using "if and only if".			
8.			

9. Write a conditional statement that is biconditional.

10. Write a conditional statement that is **not** biconditional.

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Determine if each statement is true. Rewrite each false statement using the word "some."

	Statement	True or False	Re-write
1.	All odd numbers are composite.		
2.	All points on a line are collinear.		
3.	All even numbers are divisible by 2.		

Write each statement as an if-then statement.

	Statement	If-Then
4.	All odd numbers are composite.	
5.	All points on a line are collinear.	
6.	All even numbers are divisible by 2.	

Write the negation of each statement.

	Statement	Negation
7.	$\triangle TAH$ is an isosceles triangle.	
8.	$\angle BCD$ and $\angle TYG$ are complementary angles.	
9.	$\angle ICH$ and $\angle JNV$ are adjacent angles.	
10.	$m\angle ABC + m\angle ZYX = 97°$	

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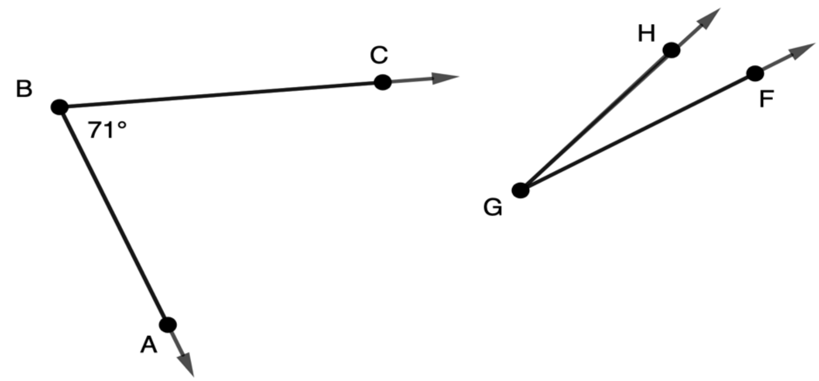
Complete the two-column proof.

Given: $\angle ABC$ and $\angle HGF$ are complementary; $m\angle ABC = 71°$

Prove: $m\angle HGF = 19°$



	Statement	Reason
1.	$\angle ABC$ and $\angle HGF$ are complementary	
2.		Definition of complementary angles
3.	$m\angle ABC = 71°$	
4.		Substitution Property
5.	$m\angle HGF = 19°$	

Complete the paragraph proof to answer questions 6-10.

Given: $\angle 1$ and $\angle 2$ are supplementary and $\angle 2$ and $\angle 3$ are complementary.

Prove: $\angle 1 \cong \angle 3$

It is given that 6. ______________________________________ and $\angle 2$ and $\angle 3$ are complementary. By the definition of complementary angles, 7. ___________________ and $m\angle 2 + m\angle 3 = 90°$. By the 8. _________________ property, $m\angle 1 + m\angle 2 = m\angle 2 + m\angle 3$. Then, $m\angle 1 =$ 9. _____________ by the subtraction property. Therefore, $\angle 1 \cong \angle 3$ by the 10. __________________________________.

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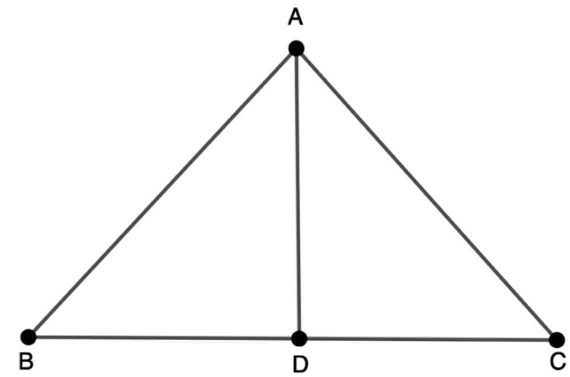
Complete the two-column proof.

Given: $\angle ADB$ is a right angle, $\angle ADB$ and $\angle ADC$ are supplementary.

Prove: $\angle ADB \cong \angle ADC$

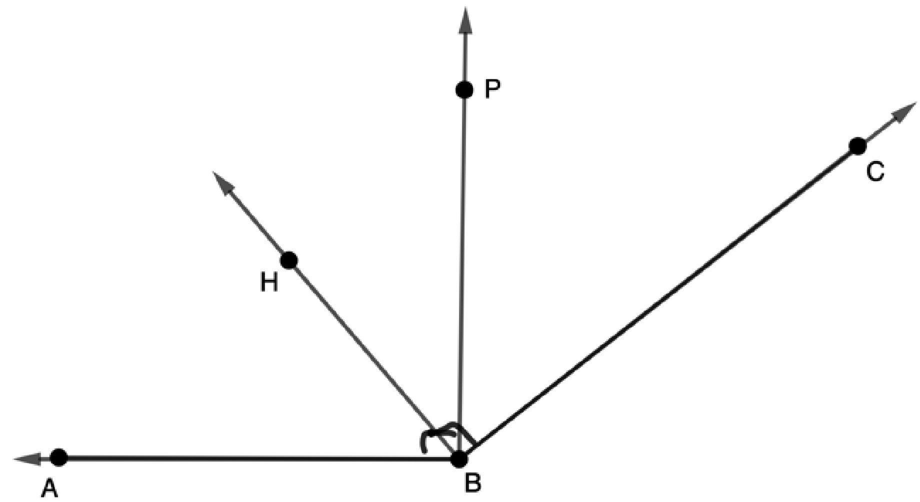


Statement	**Reason**
$\angle ADB$ is a right angle	Given
1.	Definition of right angle
$\angle ADB$ and $\angle ADC$ are supplementary	Given
$m\angle ADC + m\angle ADB = 180°$	2.
3.	Substitution Property
$m\angle ADC = 90°$	Subtraction Property
4.	Transitive Property of Equality
$\angle ADC \cong \angle ADB$	5.

Complete the paragraph proof to answer questions 5-10.

Given: $\angle PBA$ and $\angle HBC$ are right angles

Prove: $\angle ABH \cong \angle PBC$



It is given that $\angle PBA$ and $\angle HBC$ are right angles. Using the addition postulate,

6. ______________________________ and 7. _______________________________.

Using the 8. _________________ property of equality, we know

$m\angle ABH + m\angle HBP = m\angle HBP + m\angle PBC$. Next, we apply the subtraction property of equality

to determine 9. ______________________. By the definition of 10. _____________,

$\angle ABH \cong \angle PBC$.